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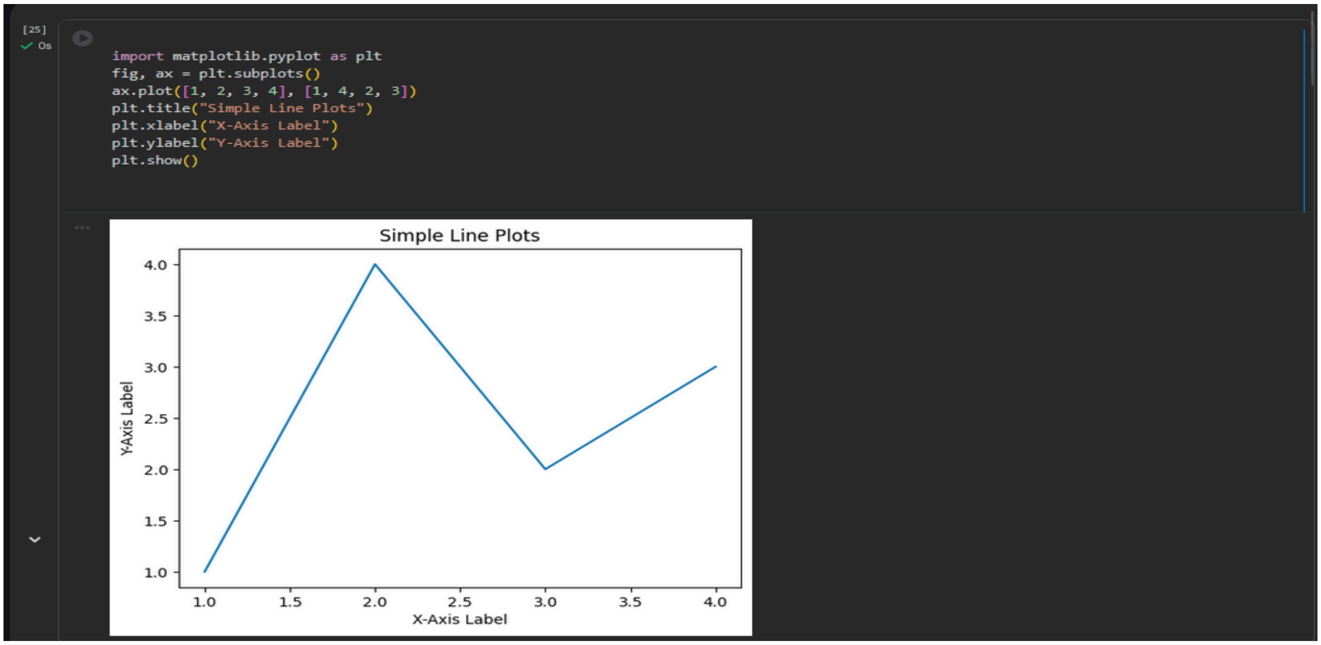
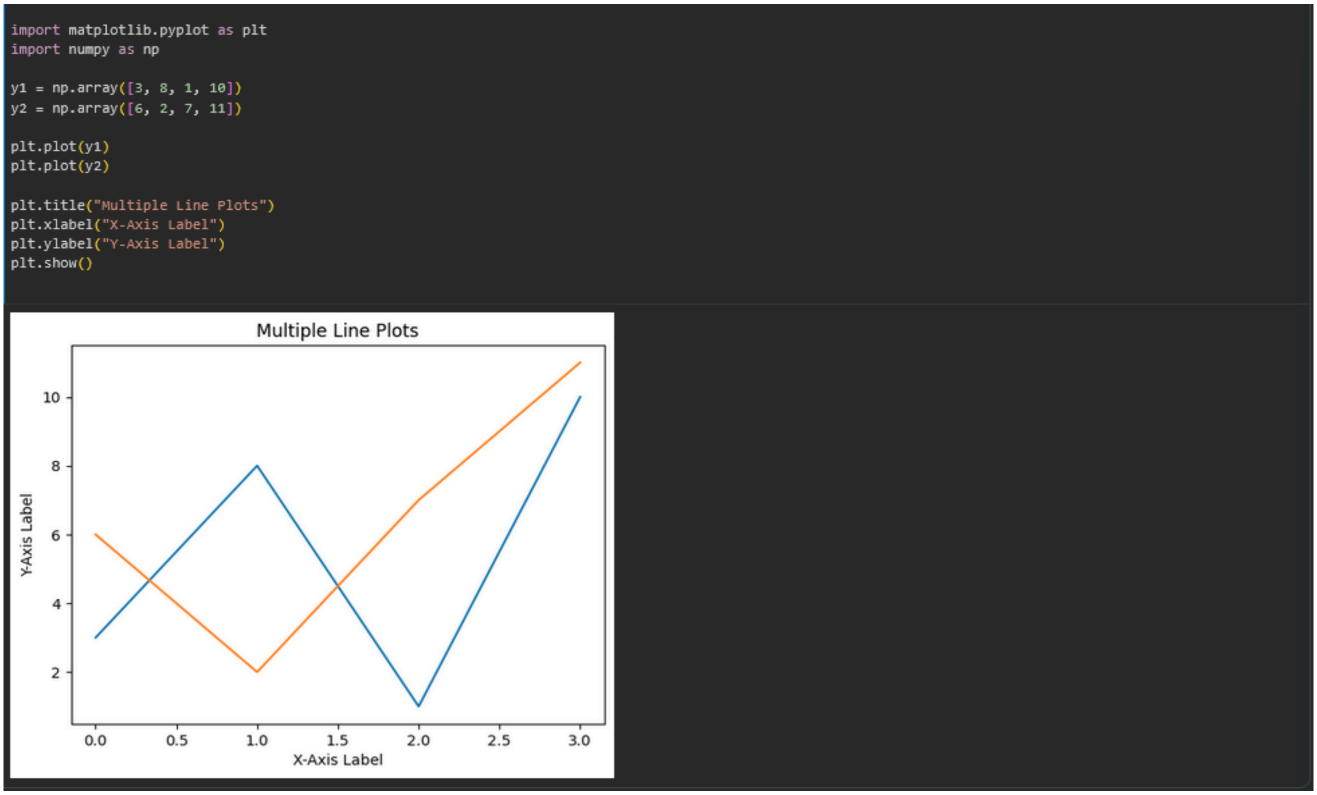
303I

1. Simple Line Plots



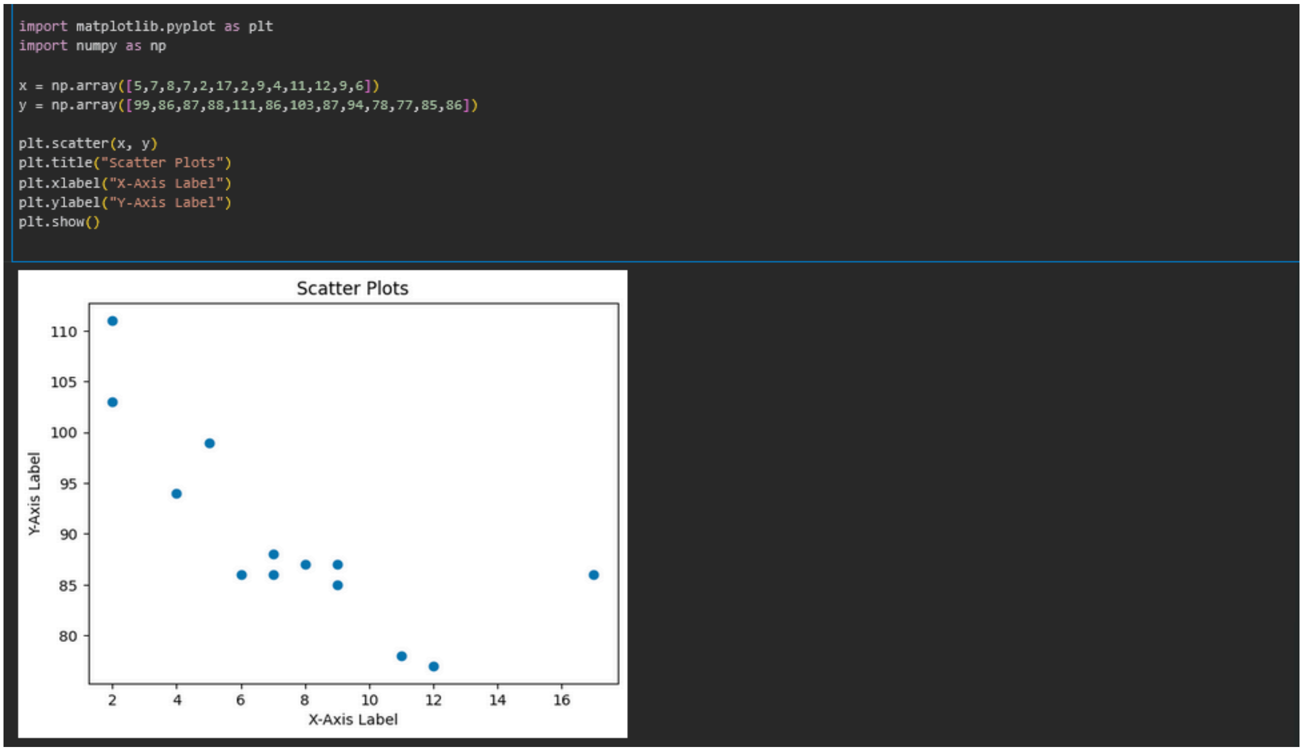
```
import matplotlib.pyplot as plt
fig, ax = plt.subplots()
ax.plot([1, 2, 3, 4], [1, 4, 2, 3])
plt.title("Simple Line Plots")
plt.xlabel("X-Axis Label")
plt.ylabel("Y-Axis Label")
plt.show()
```

2. Multiple Line Plots



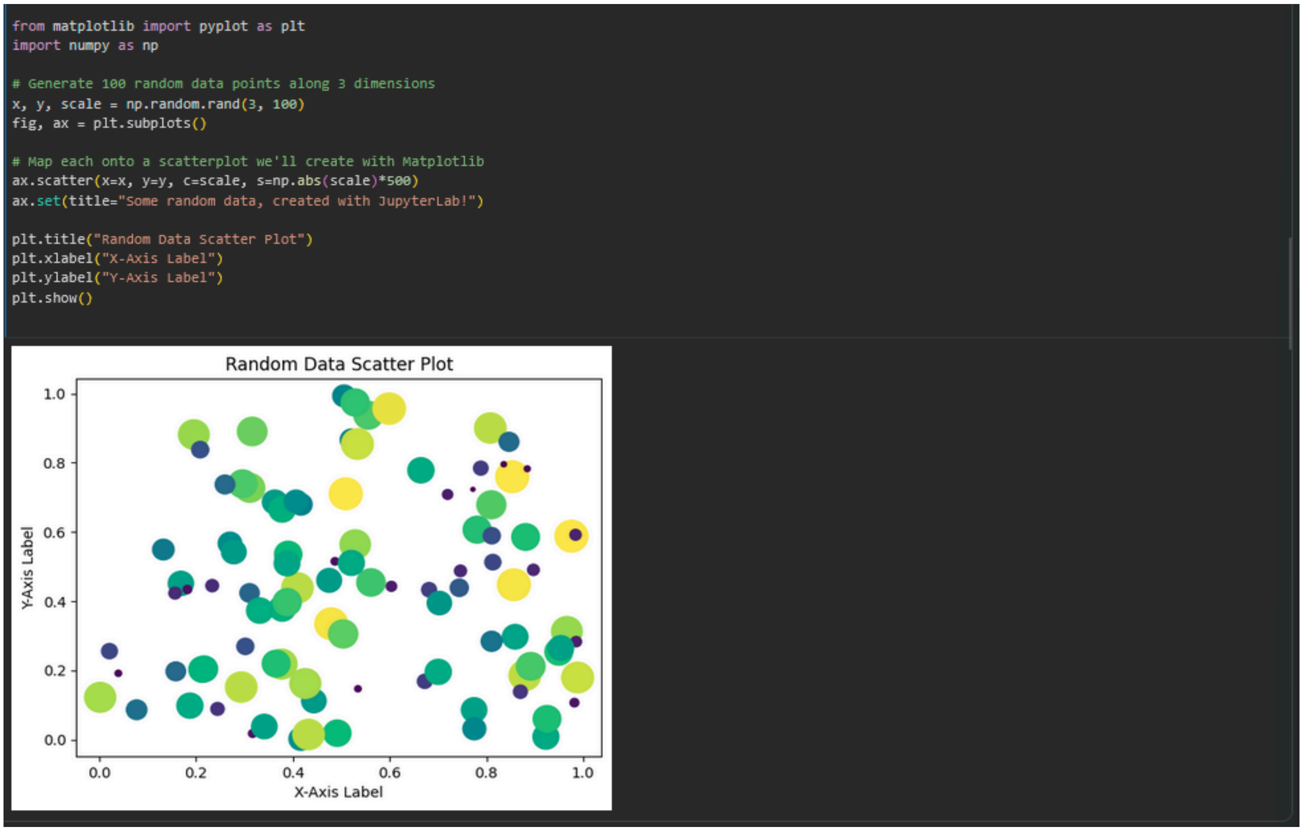
```
import matplotlib.pyplot as plt
import numpy as np

y1 = np.array([3, 8, 1, 10])
y2 = np.array([6, 2, 7, 11])

plt.plot(y1)
plt.plot(y2)

plt.title("Multiple Line Plots")
plt.xlabel("X-Axis Label")
plt.ylabel("Y-Axis Label")
plt.show()
```

3. Scatter Plots



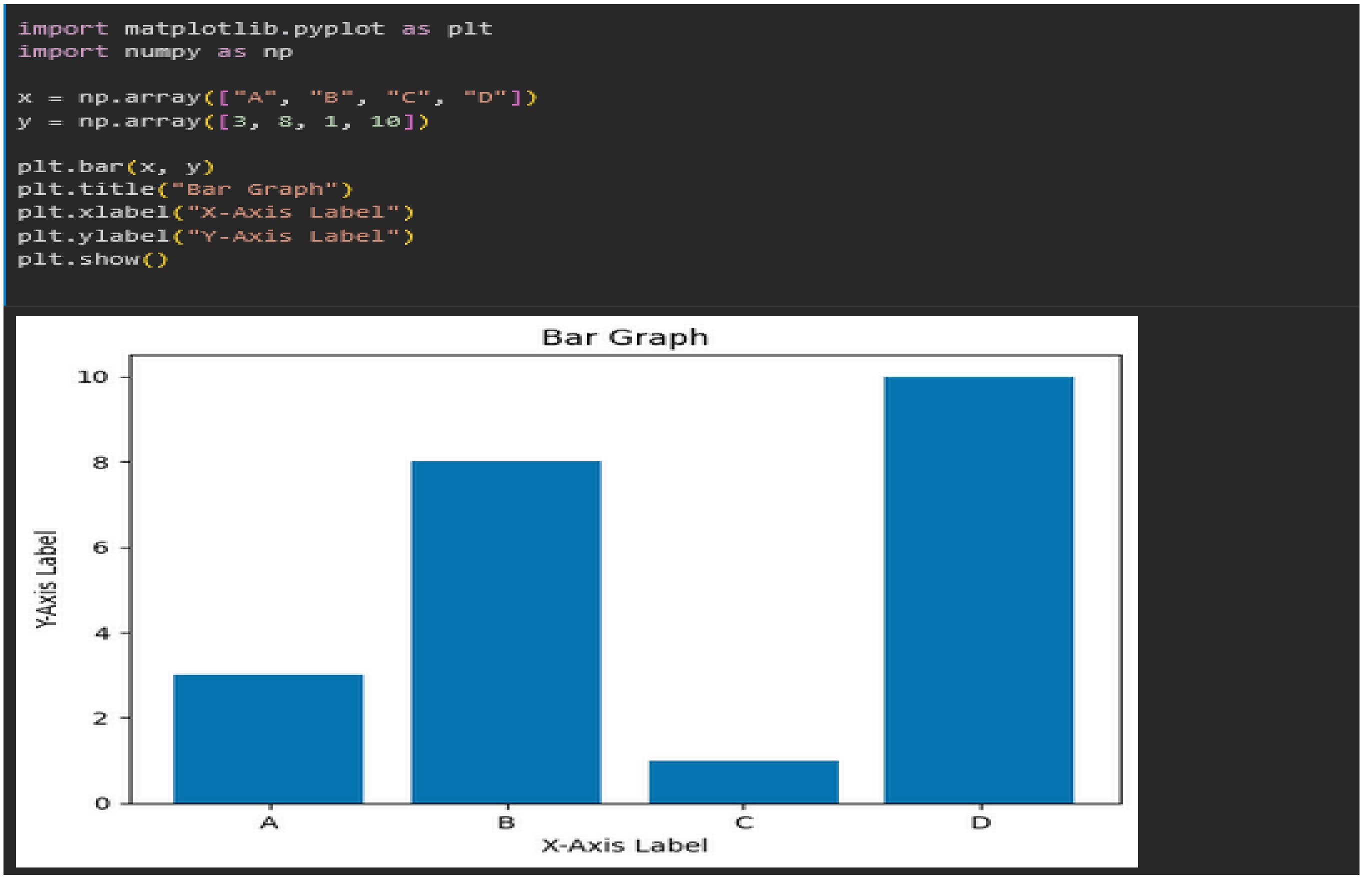
```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([5,7,8,7,2,17,2,9,4,11,12,9,6])
y = np.array([99,86,87,88,111,86,103,87,94,78,77,85,86])

plt.scatter(x, y)
plt.title("Scatter Plots")
plt.xlabel("X-Axis Label")
plt.ylabel("Y-Axis Label")
plt.show()
```

4. Random Data Scatter Plot



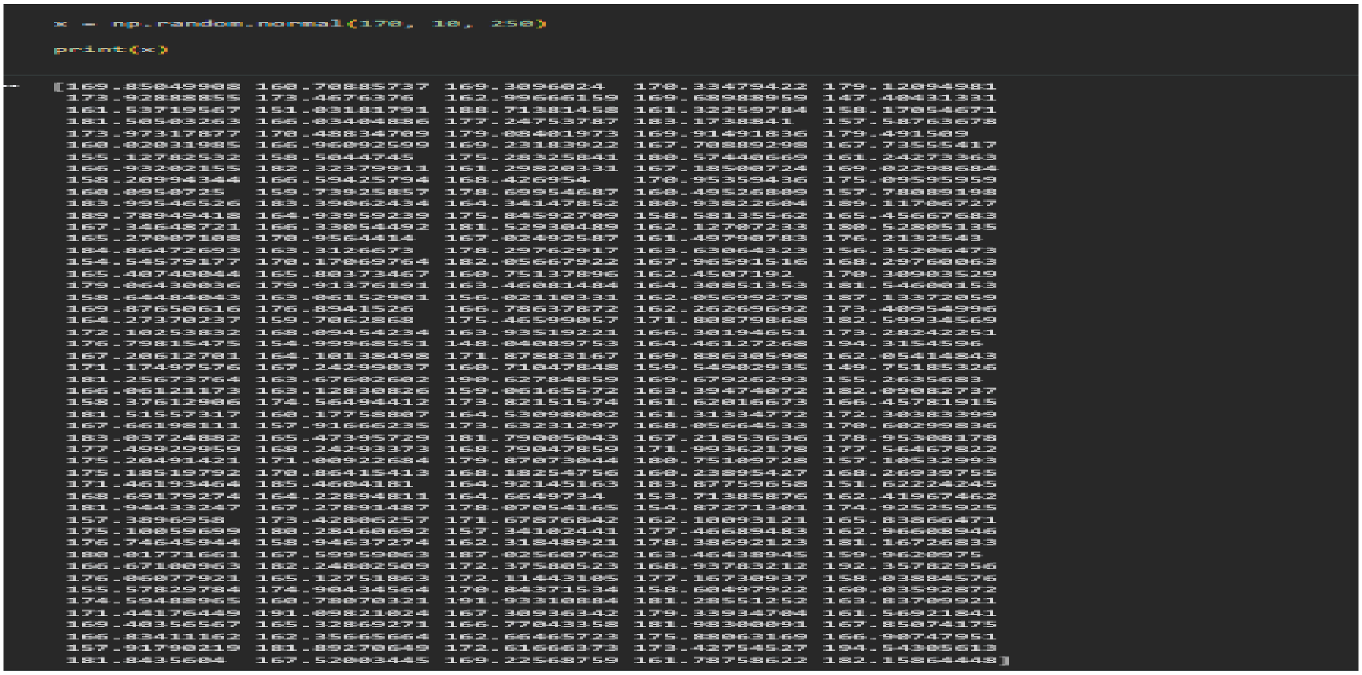
```
from matplotlib import pyplot as plt
import numpy as np

# Generate 100 random data points along 3 dimensions
x, y, scale = np.random.rand(3, 100)
fig, ax = plt.subplots()

# Map each onto a scatterplot we'll create with Matplotlib
ax.scatter(x=x, y=y, c=scale, s=np.abs(scale)*500)
ax.set(title="Some random data, created with JupyterLab!")

plt.title("Random Data Scatter Plot")
plt.xlabel("X-Axis Label")
plt.ylabel("Y-Axis Label")
plt.show()
```

5. Bar Graph



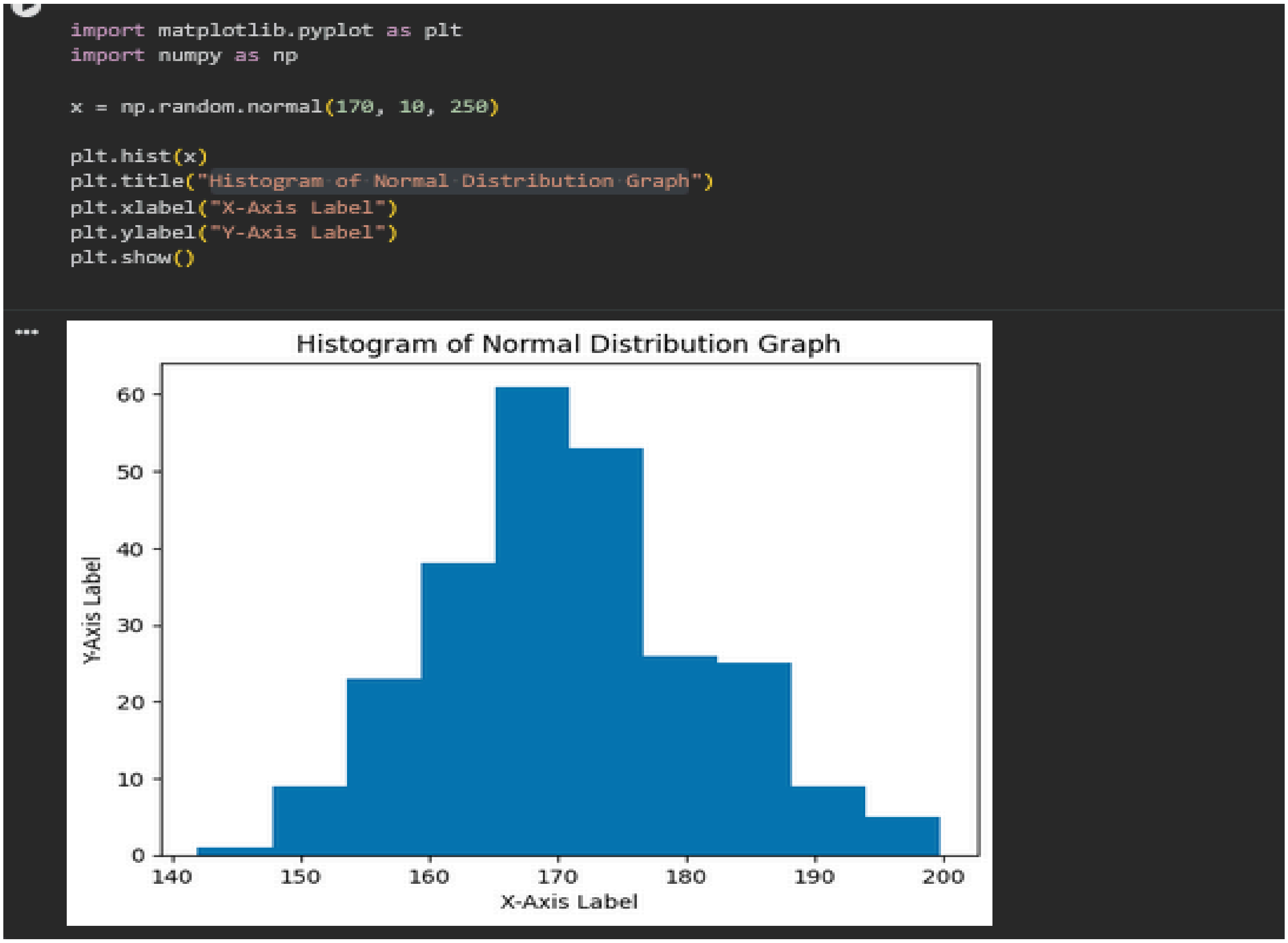
```python
import matplotlib.pyplot as plt
import numpy as np

x = np.array(["A", "B", "C", "D"])
y = np.array([3, 8, 1, 10])

plt.bar(x, y)
plt.title("Bar Graph")
plt.xlabel("X-Axis Label")
plt.ylabel("Y-Axis Label")
plt.show()
```

6. Random Numbers



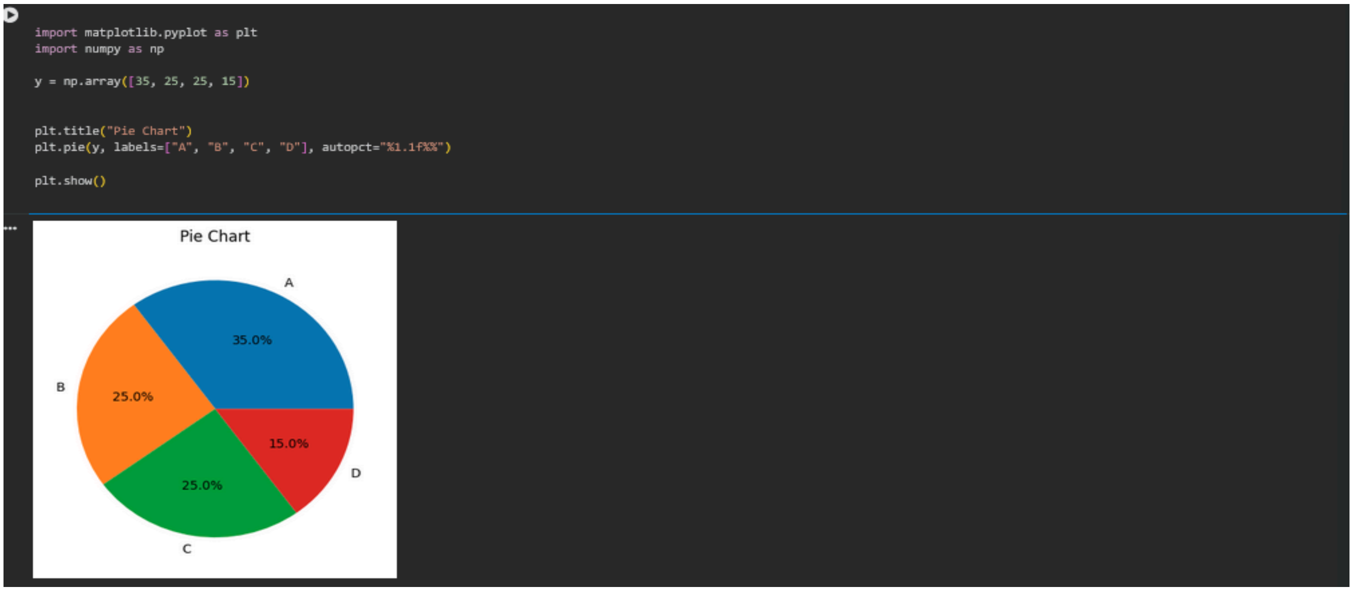
```python
x = np.random.normal(170, 10, 250)
print(x)
```

7. Histogram of Normal Distributions Graph



```
import matplotlib.pyplot as plt
import numpy as np

x = np.random.normal(170, 10, 250)

plt.hist(x)
plt.title("Histogram of Normal Distribution Graph")
plt.xlabel("X-Axis Label")
plt.ylabel("Y-Axis Label")
plt.show()
```

8. Pie Charts



```
import matplotlib.pyplot as plt
import numpy as np

y = np.array([35, 25, 25, 15])


plt.title("Pie Chart")
plt.pie(y, labels=["A", "B", "C", "D"], autopct="%1.1f%%")

plt.show()
```